A bit of mathematical terminology

A **Set** is a collection of objects.

Example

C=The set of Canadian celebrities
C = {Wayne Gretzky, Celine Dion,Justin Bieber,...}

But we are mainly interested in sets containing mathematical objects (e.g., numbers, vectors, matrices,...)

A **Subset** $\mathcal{T}$ of a set $\mathcal{S}$ is another set which contains some of elements of the set $\mathcal{S}$.

To indicate that $\mathcal{T}$ is a subset of $\mathcal{S}$, we usually write $\mathcal{T} \subseteq \mathcal{S}$.

Example

If $\mathcal{F}$ = the set of female Canadian celebrities. Then $\mathcal{F} \subseteq C$.

Sets in Mathematics

Certain sets are important for this course.

The empty set

The set with no elements is called the *empty set*, and mathematicians usually denote it by a funky notation, e.g.,

$$\varnothing \quad \text{or} \quad \{\}.$$

The empty set is a subset of *every* set.

Some examples of mathematical sets

Example

The set of integers is denoted by $\mathbb{Z}$. Recall that:

$$\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$$

The set of real numbers is denoted by $\mathbb{R}$.

$\diamond$ A real number is any number you can imagine which has a decimal representation. For example, $-3 \in \mathbb{R}$, $1 - 2\sqrt{2} \in \mathbb{R}$, $1.3333333\cdots \in \mathbb{R}$,...